

# The Framing Transformer: Where the Spin-1/2 Content of the Möbius-Screw Actually Lives

Companion computation to *The Möbius-Screw Electron*

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## Abstract

The Möbius-screw note models the electron as a framed unknot with self-linking number  $Sl = 2$  under the torus framing, and identifies  $Sl \leftrightarrow g$  as a bookkeeping statement. This note computes the full chain of maps from the centerline to the  $SU(2)$  lift of its frame field, and evaluates every stage. Three results. (i) The Călugăreanu decomposition is confirmed numerically:  $Tw = -1.0338$ ,  $Wr = -0.9662$ ,  $Tw + Wr = -2.000000$ , and the Gauss linking integral of the centerline with its framing pushoff returns  $-2.000000$ ; the magnitude  $|Sl| = pq = 2$  is as stated, the sign records that the embedding as parameterised is left-handed. (ii) The frame loop is genuinely *spinorial*: its class in  $\pi_1(SO(3)) \cong \mathbb{Z}/2$  is non-trivial, so the  $SU(2)$  lift does not close after one circuit. This is proved in closed form, not merely computed. (iii) The spin content is carried by a *parity bit*, not by the value 2. For torus curves the holonomy is  $\sigma = (-1)^{p+q} = (-1)^{Sl+1}$ , so  $Sl = 2$  and  $Sl = 0$  lie in the same class: an untwisted round circle is spinorial in exactly the same sense. The magnitude of  $Sl$  carries no spin information, which withdraws the support the  $Sl = 2 \leftrightarrow g = 2$  identification was taken to have. What survives is sharper and smaller: the double cover in this model comes from the meridian half-angle  $\varphi/2$  in the parameterisation, i.e. from the *odd* meridian winding  $q = 1$ . The parity law itself is not new — it is standard in the framed-curve and Kirby-calculus literature (§9) — and the contribution here is its evaluation on this curve, together with the resulting correction.

**Publication status (24 September 2026).** This version includes a claim-specific amendment at the end of the main text. That amendment governs earlier physical interpretations and distinguishes retained calculations from withdrawn or unresolved inferences.

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# 1 Scope

This note is a computation, not a new model. It takes the centerline and framing already defined in the Möbius-screw note and asks a single question: *at which stage, if any, does spin-1/2 enter?* Every number below is produced by the three scripts `code/framed_unknot/framing_transformer.py` (Sections 2–5), `code/framed_unknot/moment_ratio.py` (Section 6), and `code/framed_unknot/one_object.py` (Section 7), archived in `docs/framed_unknot_results.json`, `docs/framed_unknot_moment_ratio.json`, and `docs/framed_unknot_one_object.json` respectively.

Nothing here bears on QED, on radiative corrections to  $g - 2$ , or on the capacitance estimate of  $\alpha$ . The result is topological.

## 2 The transformer chain

The object is the closed curve,  $\varphi \in [0, 4\pi)$ ,

$$\gamma(\varphi) = \begin{pmatrix} (R + a \cos(\varphi/2)) \cos \varphi \\ (R + a \cos(\varphi/2)) \sin \varphi \\ a \sin(\varphi/2) \end{pmatrix}, \quad (1)$$

which is the  $(p, q) = (2, 1)$  curve on the torus of radii  $(R, a)$ : longitude angle  $\theta = \varphi$ , meridian angle  $\psi = \varphi/2$ . The torus surface normal

$$\mathbf{U}(\varphi) = (\cos(\varphi/2) \cos \varphi, \cos(\varphi/2) \sin \varphi, \sin(\varphi/2)) \quad (2)$$

is the natural framing. Because  $\gamma$  lies in the surface whose normal is  $\mathbf{U}$ , orthogonality  $\mathbf{T} \cdot \mathbf{U} = 0$  holds identically (verified to  $2.2 \times 10^{-16}$ ).

The chain from geometry to spin is

$$\gamma \xrightarrow{\text{framing}} \mathbf{U} \xrightarrow{\text{CWF}} Sl = Tw + Wr \xrightarrow{\text{adapted frame}} F : S^1 \rightarrow SO(3) \xrightarrow{\text{lift}} q : S^1 \rightarrow SU(2),$$

with  $F(\varphi) = [\mathbf{T}, \mathbf{U}, \mathbf{T} \times \mathbf{U}]$ . The last arrow is the only stage at which a double cover can appear, and whether it succeeds is decided by the single sign

$$\sigma = \frac{q(4\pi)}{q(0)} \in \{+1, -1\}, \quad (3)$$

$\sigma = +1$  meaning the lift closes (frame returns after one circuit) and  $\sigma = -1$  meaning it does not (two circuits required).

## 3 Stage 1: the Călugăreanu decomposition

Twist is evaluated as  $Tw = \frac{1}{2\pi} \oint (\mathbf{T} \times \mathbf{U}) \cdot \mathbf{U}' d\varphi$ , writhe by the Gauss double integral, and  $Sl$  independently as the linking number  $Lk(\gamma, \gamma + \epsilon \mathbf{U})$ .

| Quantity                               | Value     | Method                 |
|----------------------------------------|-----------|------------------------|
| $Tw$                                   | −1.033761 | twist integral         |
| $Wr$                                   | −0.966239 | Gauss double integral  |
| $Tw + Wr$                              | −2.000000 | Călugăreanu sum        |
| $Lk(\gamma, \gamma + 0.10a\mathbf{U})$ | −2.000000 | Gauss linking integral |
| $Lk(\gamma, \gamma + 0.05a\mathbf{U})$ | −2.000019 | same, halved offset    |
| $pq$                                   | 2         | torus-framing formula  |

The magnitude  $|Sl| = 2$  claimed in the Möbius-screw note is confirmed by two independent routes. The negative sign is not an error: it records that the embedding (1), with meridian angle advancing as  $+\varphi/2$ , is left-handed under the standard Gauss convention. Only  $|Sl|$  and its parity are used below.

**Remark 1** (What is and is not topological here).  $Tw$  and  $Wr$  individually depend on  $R$  and  $a$  and are not invariants; only their sum is. The near-equal split  $(-1.03, -0.97)$  at  $a/R = 0.3$  is a coincidence of that aspect ratio, not a structural fact, and should not be read as “one unit of twist plus one unit of writhe.”

## 4 Stage 2: the $SU(2)$ lift, in closed form

The framing (2) is exactly the sphere parameterisation at longitude  $\varphi$  and latitude  $\varphi/2$ , so  $\mathbf{U}(\varphi) = A(\varphi) \mathbf{e}_x$  with

$$A(\varphi) = R_z(\varphi) R_y(-\varphi/2). \quad (4)$$

Lifting each factor to unit quaternions ( $R_z(t) \mapsto \cos \frac{t}{2} + k \sin \frac{t}{2}$ ,  $R_y(t) \mapsto \cos \frac{t}{2} + j \sin \frac{t}{2}$ ) gives the continuous lift with  $q(0) = 1$ ,

$$q(\varphi) = \left[ \cos \frac{\varphi}{2} + k \sin \frac{\varphi}{2} \right] \left[ \cos \frac{\varphi}{4} - j \sin \frac{\varphi}{4} \right]. \quad (5)$$

**Proposition 1** (Spinorial holonomy). *For the curve (1) with framing (2),*

$$q(4\pi) = [\cos 2\pi + k \sin 2\pi] [\cos \pi - j \sin \pi] = (1)(-1) = -1,$$

so  $\sigma = -1$ : the frame loop represents the non-trivial class of  $\pi_1(SO(3)) \cong \mathbb{Z}/2$  and its  $SU(2)$  lift does not close after one circuit of the curve.

Where the sign comes from. The meridian factor. Over the closed curve the meridian angle advances by  $2\pi$ , so its half-angle in (5) advances by only  $\pi$ , landing on  $-1$ . The longitude factor advances by  $4\pi$ , half-angle  $2\pi$ , landing on  $+1$  and contributing nothing.  $\square$

Proposition 1 is confirmed numerically three ways, all returning  $\sigma = -1.000000$ : the adapted frame  $[\mathbf{T}, \mathbf{U}, \mathbf{T} \times \mathbf{U}]$  lifted by continuity, the framing field  $A(\varphi)$  lifted by continuity, and direct evaluation of (5).

## 5 Stage 3: the value 2 does no work

Generalising Proposition 1 to the  $(p, q)$  torus curve with torus framing: the longitude half-angle advances by  $\pi p$  and the meridian half-angle by  $\pi q$ , so

$$\sigma = (-1)^{p+q}. \quad (6)$$

Since  $Sl = pq$  and  $\gcd(p, q) = 1$  forbids both being even,  $p + q$  is odd exactly when  $pq$  is even, hence

$$\sigma = (-1)^{Sl+1}. \quad (7)$$

**Remark 2** (Convention, stated so it cannot be misread). Equation (7) is normalised to the adapted frame  $[\mathbf{T}, \mathbf{U}, \mathbf{T} \times \mathbf{U}]$  with  $Sl$  measured against the Seifert (0-) framing. In that normalisation the round unknot with the 0-framing is the generator of  $\pi_1(SO(3))$  — it does not lift — because the tangent indicatrix itself contributes one turn. Absolute parity statements are therefore convention-dependent and easy to get backwards. The convention-free form, which is what should be quoted, is:

Incrementing the framing by 1 flips the  $\pi_1(SO(3))$  class; framings differing by 2 lie in the same component.

This is the belt trick, and it is the statement that survives any choice of reference framing. Compare Needham [7], Thm. 3.7, where an  $|h|$ -fold covered circle of self-linking  $k$  has closed quaternion lift iff  $h + k$  is even — which for  $h = 1$  reproduces the control column below exactly.

This was checked against an independent control family — round circles carrying  $n$  framing twists, for which  $Wr = 0$  and  $|Sl| = n$  — with the measured result reproducing (7) exactly:

| $n$ | $Tw$ | $Wr$ | $Sl$ | class     |
|-----|------|------|------|-----------|
| -2  | +2   | 0    | +2   | spinorial |
| -1  | +1   | 0    | +1   | trivial   |
| 0   | 0    | 0    | 0    | spinorial |
| 1   | -1   | 0    | -1   | trivial   |
| 2   | -2   | 0    | -2   | spinorial |
| 3   | -3   | 0    | -3   | trivial   |
| 4   | -4   | 0    | -4   | spinorial |

**Corollary 1** (The identification  $Sl = 2 \leftrightarrow g = 2$  loses its support). *The spin-relevant content of the self-linking number is one bit, its parity. A framed loop with  $Sl = 0$  — an ordinary round circle with an untwisted framing — is spinorial in precisely the same sense as the Möbius screw with  $|Sl| = 2$ . The value 2 therefore carries no spin information that 0 does not also carry, and cannot be what produces the double cover.*

**Remark 3** (Status change). *The Möbius-screw note (§3.3) writes that  $Sl = 2$  “matches the  $4\pi$  (two-turn) return of a spin-1/2 frame,” and labels this a model identification rather than a derivation. Corollary 1 removes the geometric motivation for the identification itself: the matching quantity is a  $\mathbb{Z}/2$  class, and 2 and 0 are the same class. Under the repository’s tier discipline this is a **T4 (explicitly falsified)** outcome for the reading of  $Sl = 2$  as the geometric origin of  $g = 2$ , and should be recorded in the Elimination Ledger. The rest of the Möbius-screw note is untouched: the geometry, the winding numbers, and the  $|Sl| = 2$  computation are all confirmed.*

## 6 What survives

Something real remains, and it is more specific than what was claimed. The double cover in this model is produced by the **meridian half-angle**  $\varphi/2$  appearing in (1) and (2) — that is, by the *odd meridian winding*  $q = 1$ , via (6). The screw’s defining feature, that the frame must traverse the curve twice to return, is a genuine topological property of the shape, proved in Proposition 1, and it is exactly the  $720^\circ$  property of a spin-1/2 frame.

So the model does land on the spinorial side of a genuine  $\mathbb{Z}/2$  invariant. That is a legitimate structural result and it is what the tornado/Archimedean-screw picture was reaching for. It is also considerably weaker than a derivation of  $g$ : a  $\mathbb{Z}/2$  bit cannot produce a real number, and any model claiming a magnitude from this structure needs a mechanism that (6) does not supply.

### 6.1 The tornado picture is the same object, not a metaphor

Appendix A of the Möbius-screw note offers a “tornado” reading — a funnel narrowing to a high-torsion throat around a quiet axis — and labels it interpretive, adding no formal claims. That labelling is more conservative than it needs to be. Sweeping the aspect ratio  $a/R$  towards 1 carries the curve continuously from a fat-hole torus to exactly that funnel: since  $\rho(\varphi) = R + a \cos(\varphi/2)$ , the closest approach to the axis is  $\rho_{\min} = R - a$ , attained at  $\varphi = 2\pi$ , and it shrinks to zero as  $a \rightarrow R$ . The throat is not an added feature of the picture; it is where the centerline meets the axis.

Across that entire sweep the invariants do not move:

| $a/R$ | $\rho_{\min}$ | $Tw$     | $Wr$     | $\sigma$ |
|-------|---------------|----------|----------|----------|
| 0.30  | 0.700         | -1.03376 | -0.96624 | -1       |
| 0.50  | 0.500         | -1.09066 | -0.90934 | -1       |
| 0.70  | 0.300         | -1.14910 | -0.85090 | -1       |
| 0.85  | 0.150         | -1.15375 | -0.84625 | -1       |
| 0.97  | 0.030         | -1.11872 | -0.88128 | -1       |
| 0.999 | 0.001         | -1.10597 | -0.89403 | -1       |

$Tw + Wr = -2.000000$  at every row, and the holonomy stays  $\sigma = -1$  throughout. This is isotopy invariance exhibited directly: the deformation is continuous and embedded for all  $a < R$ , degenerating only in the limit, so donut and vortex are one framed loop and the topology cannot distinguish them. Note also that  $Tw$  is *non-monotonic* across the sweep, turning over near  $a/R \approx 0.85$  — further evidence that the individual terms are geometry with no invariant meaning.

The consequence cuts both ways. The tornado picture is licensed as a faithful drawing of the formal object, which it was not previously claimed to be. But it inherits the object’s limits exactly: the throat is not an extra structure that could supply a magnitude, because it is reachable by deformation from a shape that has no throat at all.

## 7 The successor test, run: geometry gives $g = 1$

The obvious next step is not a larger analogy but a smaller question: compute the shape’s *magnetic moment* against its *angular momentum* and read off a  $g$  that has dimensions and can fail. We ran it (`code/framed_unknot/moment_ratio.py`). It fails, and the way it fails is instructive.

Let a particle of charge  $q$  and mass  $m$  traverse the closed centerline once per period  $T$ , and write the vector area

$$\mathbf{A} = \frac{1}{2} \oint \mathbf{r} \times d\mathbf{l}. \quad (8)$$

Then the current is  $I = q/T$ , and

$$\boldsymbol{\mu} = I\mathbf{A} = \frac{q}{T}\mathbf{A}, \quad \langle \mathbf{L} \rangle = \frac{m}{T} \oint \mathbf{r} \times d\mathbf{r} = \frac{2m}{T}\mathbf{A}. \quad (9)$$

Both are proportional to the *same* geometric factor, so

$$\frac{\boldsymbol{\mu}}{\langle \mathbf{L} \rangle} = \frac{q}{2m} \implies g = 1 \quad \text{exactly, for every closed curve.} \quad (10)$$

| Curve                             | $A_z$    | $A_z/\pi$ |
|-----------------------------------|----------|-----------|
| Möbius screw (2, 1), $a/R = 0.30$ | +6.56593 | +2.09000  |
| Möbius screw (2, 1), $a/R = 0.70$ | +7.82257 | +2.49000  |
| Möbius screw (2, 1), $a/R = 0.97$ | +9.23911 | +2.94090  |
| round circle, one turn            | +3.14159 | +1.00000  |
| round circle, two turns           | +6.28319 | +2.00000  |

The double winding is entirely real: the screw carries  $2.09\times$  the vector area of a single loop. It is also useless for  $g$ , because that factor enters  $\boldsymbol{\mu}$  and  $\langle \mathbf{L} \rangle$  identically and cancels — the same way the 2 in  $Sl$  cancelled down to a parity bit in §5. Note too that  $A_z$  is *not* an invariant: it runs from  $2.09\pi$  to  $2.94\pi$  across the throat sweep of §6.1, unlike  $Sl$ , which does not move at all.

**Corollary 2** (No-go for geometric  $g$ ). *No model in which charge and mass circulate together with a uniform  $q/m$  ratio can produce  $g \neq 1$  by geometry, whatever the winding, framing, twist, or throat. Equation (10) is the classical orbital  $g$ -factor, and it is shape-independent. Obtaining  $g \neq 1$  requires decoupling the charge and mass distributions — a specific structural requirement, and a far better-posed target than looking for a 2 in the geometry.*

## 7.1 On the dynamo reading

A natural physical reading of this shape is a dynamo, and one part of that instinct is exactly right — it is a theorem, not an analogy. For a thin magnetic flux tube of flux  $\Phi$ , the magnetic helicity is

$$H = \int \mathbf{A} \cdot \mathbf{B} d^3x = \Phi^2 (Tw + Wr) = \Phi^2 Sl, \quad (11)$$

so the Călugăreanu quantity computed in §3 *is* the magnetic helicity of a flux tube with this geometry (Moffatt [4]; Moffatt–Ricca [5]). Dynamo action is likewise about exactly the  $Tw \leftrightarrow Wr$  trade seen in §3: the stretch–twist–fold mechanism converts writhe into twist while doubling flux. The bookkeeping is the same bookkeeping.

Two things block the reading as an electron model, and neither is topological.

1. **A dynamo grows.** It has a growth rate and amplifies field by consuming kinetic energy from a flow. A stable particle is a stationary state — no growth, no energy source, no dissipation. The dynamical class is wrong.
2. **Helicity is the invariant we already have.** Since  $H = \Phi^2 Sl$ , the dynamo framing supplies the same integer, in the same units of  $\Phi^2$ . It adds physical content but no new magnitude, and Corollary 2 still applies.

The nearest object that *is* stationary is the force-free or Taylor state: by Woltjer’s theorem, a plasma relaxing at fixed magnetic helicity minimises energy at  $\nabla \times \mathbf{B} = \lambda \mathbf{B}$ , a self-organised helicity-carrying equilibrium (a spheromak). That is the right dynamical class, and  $\lambda$  carries dimensions of inverse length, so unlike the parity bit it does set a scale. It still does not evade Corollary 2 on its own — a force-free state with co-located charge and mass gives  $g = 1$  like everything else — but it poses the question in the only form that could survive: what, in this geometry, separates where the charge sits from where the mass sits?

## 8 One object, four axes — and the Laplace eigenspace

Three structural readings of this geometry turn out to be exactly right, and together they relocate where spin-1/2 lives. All are verified in `code/framed_unknot/one_object.py`.

### 8.1 $(2, 1)$ is the $xy$ -axis twisted

The pair  $(p, q) = (2, 1)$  is not two independent windings. It is two Euler angles of a *single* rotation, locked at a 2:1 rate:

$$U(\varphi) = A(\varphi) \mathbf{e}_x, \quad A(\varphi) = R_z(\varphi) R_y(-\varphi/2), \quad (12)$$

so  $p = 2$  is the azimuthal turn in the  $xy$ -plane and  $q = 1$  is the tilt. Checked against the direct torus normal, the difference is identically zero (0 to machine precision) — not merely small. This is why the lift factorises the way it does in (5), and why the sign is carried entirely by the half-angle of the tilt.

### 8.2 The framed curve is one quaternion curve

A framed curve looks like  $3 + 1$  pieces of data — a centerline plus a normal field — only in the naive presentation. Through the frame-Hopf map it is a *single* quaternion-valued curve  $q(\varphi)$  on  $S^3$ , from which the framing and the remaining frame leg are recovered by conjugation alone:

$$\mathbf{U} = q \mathbf{e}_x \bar{q}, \quad \mathbf{V} = q \mathbf{e}_z \bar{q}. \quad (13)$$

Reconstructing from  $q$  and nothing else reproduces the direct formulas to  $3.3 \times 10^{-16}$ , with  $|q| = 1$  to  $2.2 \times 10^{-16}$ . Four numbers, one object. (This is Needham’s construction, [7] §2.)

### 8.3 Laplace: where spin-1/2 actually enters

The four quaternion coordinates are not an arbitrary basis. Restricted to  $S^3 = SU(2)$  they are exactly the first nonzero eigenspace of the Laplace–Beltrami operator:

$$\Delta_{S^3} x_i = -3 x_i, \quad \text{degeneracy } (k+1)^2 = 4 \text{ at } k = 1, \quad (14)$$

measured here as  $-3.00000$  on each of the four axes (max error  $8 \times 10^{-7}$  by finite differences on  $2 \times 10^4$  sampled points). Under  $SU(2)_L \times SU(2)_R$  that eigenspace is the  $(\frac{1}{2}, \frac{1}{2})$  representation.

**Remark 4** (The relocation). *This is the honest home of spin-1/2 in this picture, and it is not the self-linking number. Spin enters as a Laplace eigenspace on the quaternion sphere — i.e. through representation theory, which is precisely where Lévy-Leblond [13] locates  $g = 2$  as well. §5 showed the linking number contributes one parity bit; §7 showed the geometry contributes  $g = 1$ . What is left carrying the spin structure is the rep theory of  $S^3$ , which was never a geometric claim about the electron’s shape.*

### 8.4 The chain silently requires rationality

Worth recording because nothing above states it: the whole transformer presupposes a *closed* curve, and a torus curve closes iff its winding ratio  $q/p$  is rational.

| Winding ratio     | Closes after  |
|-------------------|---------------|
| 1/2 (2,1)         | 2 turns       |
| 2/3 (3,2)         | 3 turns       |
| 3/5 (5,3)         | 5 turns       |
| $1/\phi$ (golden) | never — dense |
| $1/\sqrt{2}$      | never — dense |

At irrational slope the curve never closes, fills the torus densely, and has *no* self-linking number and *no*  $\pi_1$  class at all. There is a real tension here worth naming: KAM theory makes the noble (most irrational, golden-ratio) windings the *most* robust invariant tori under perturbation, while topological quantisation — everything in this note — requires the rational ones. Maximal dynamical stability and topological quantisation pull in opposite directions on a torus. Any model that wants both owes an account of which it is choosing.

## 9 Relation to the established literature

The results of §§3–5 are not new mathematics. They are a computation of known structure applied to this specific curve, and it is worth recording exactly which parts are textbook, because that is what makes the T4 verdict binding rather than merely our opinion.

**The parity law is known.** That the space of closed framed curves in  $\mathbb{R}^3$  has exactly two path components, distinguished by the parity of the self-linking number, with the mechanism being the double cover  $SU(2) \rightarrow SO(3)$ , is stated outright by Needham [7], §2.2, in the framed-curve setting. The same fact appears in Kirby calculus as: a framing of a knot induces a spin structure on it, *even* framings giving the bounding structure and *odd* framings the non-bounding one (Gompf–Stipsicz [8], §5.6–5.7). The explicit quaternionic frame–Hopf map and the closed/anti-closed dichotomy used in §4 are Needham’s §2. For the Călugăreanu decomposition the modern geometric reference is Dennis–Hannay [6]. Our (5) and Proposition 1 are that machinery evaluated on (1); the value added is the specific evaluation, not the framework.

**Framing genuinely carries spin data — at the level of a  $\mathbb{Z}/2$ .** This is real and load-bearing elsewhere in physics. It is why Chern–Simons theory needs Atiyah’s 2-framings [9] or  $p_1$ -structures rather than framings; the current careful treatment is Freed–Teleman [10]. It is also why the topological route to *half-integer angular momentum* works at all: Finkelstein–Rubinstein [11] and, in the form closest to the present note, Wilczek–Zee [12], where a *linking number* (the Hopf term) produces fractional spin and statistics in 2+1 dimensions. So “linking number  $\rightarrow$  spin” is a real mechanism. What it produces is a phase and a statistics sector, not a gyromagnetic ratio.

**The step to  $g$  is made nowhere.** A literature sweep for a peer-reviewed derivation of  $g = 2$  from a framed loop, ribbon, or self-linking structure returns nothing in the mainstream record. Decisively in the other direction, Lévy-Leblond [13] showed that  $g = 2$  follows from *linearising the Schrödinger equation* — it is a consequence of the spinor representation, and is neither relativistic nor topological in origin. Any geometric account of  $g$  has to beat that, and none does.

The nearby speculative literature should be cited, if at all, with its standing attached. Kelvin’s vortex atoms are historical only. Battey-Pratt & Racey (1980) is published but effectively uncited outside fringe work and should not be used as authority. Schiller’s strand model appears in peer-reviewed venues but its substance is self-published with no independent uptake. Bilson-Thompson’s braided preons are peer-reviewed and have modest uptake within loop quantum gravity, but the author states explicitly that the model does *not* explain the origin of spin — ribbon twist there encodes electric charge. Hestenes’ geometric-algebra reformulation of the Dirac equation is mathematically sound; the physical zitterbewegung reading is a minority interpretation and recovers  $g = 2$  from the Dirac equation, which already supplies it.

**One trap worth naming.** “Half-integer hopfion” and “fractional skyrmion” in the 2024–2026 condensed-matter literature mean a half-integer *topological index*, not half-integer *angular momentum*. These are not spin-1/2 objects and must not be cited as precedent for one.

## Reproduction

python3 code/framed\_unknot/framing\_transformer.py

Grid  $N = 2000$  for the  $(2, 1)$  curve,  $N = 1200$  for the controls,  $R = 1$ ,  $a = 0.3$ . Results archived at docs/framed\_unknot\_results.json.

## Amendment: scope and evidence, 24 September 2026

The explicit parity correction remains in force. The later carrier audit separates it from magnetic-moment dynamics: uniform classical charge-to-mass density gives  $g = 1$  for every closed curve, while varying the densities can change the moment without changing the framing. Thus the geometric parity result neither supplies the Dirac magnetic moment nor identifies an electron. Its value is the precisely scoped lift computation and the falsification of the earlier magnitude inference.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.

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